PaymentProcessor.java

Teacher's Guide & Speaker Notes for Presentation

TEACHING THE CLASS: The Execution Engine

```
public class PaymentProcessor
```

How to explain this: Tell the audience that this is the brain of the operation. Highlight that this class does NOT extend anything. It is completely independent. Its only job is to manage the queue of payments.

TEACHING THE VARIABLES: Polymorphic Storage

```
private List<Payment> queue;
```

How to explain this: This is a huge talking point! Ask them to notice that the list holds "Payment", not "CreditCardPayment". Explain that this is Polymorphism in action. Because every payment type inherits from the Payment class, we can throw Credit Cards, PayPals, and Bank Transfers all into the exact same list, and Java treats them all as just "Payments".

TEACHING THE ALGORITHM: Dynamic Dispatch (The Climax)

```
for (Payment payment : queue) {
    boolean success = payment.process(log);
}
```

How to explain this: This is the climax of your presentation. Point out that there are NO "if/else" statements here. We don't say "If this is a credit card, do X. If it's PayPal, do Y".

Explain how it works: We just blindly call "payment.process()". The Java Virtual Machine (JVM) is smart enough to look at the object in memory, realize it is actually a CreditCard, and run the CreditCard's specific code!

Conclude by explaining why this is powerful: "If we want to add Apple Pay tomorrow, we just create an ApplePay class. We don't have to change a single line of code in this PaymentProcessor file ever again!"

TEACHING THE MATH: Java 8 Streams

```
public double totalSuccessful()
```

How to explain this: Briefly explain that this uses "Streams" to loop through the history in one elegant line of code. It filters out the failures, maps the remaining items to get their monetary amounts, and sums them up.